

Firebase Setup Guide

This guide will help you configure Firebase for your Chicken Processing Platform.

Step 1: Get Your Firebase Credentials

1. Go to [Firebase Console](#)
2. Select your project or create a new one
3. Click the gear icon  next to "Project Overview"
4. Select "Project settings"
5. Go to the "Service accounts" tab
6. Click "Generate new private key"
7. Download the JSON file

Step 2: Update Your .env File

Open the `.env` file in your project and replace these values:

Required Values:

- `FIREBASE_PROJECT_ID` = Your Firebase project ID
- `FIREBASE_PRIVATE_KEY_ID` = From the downloaded JSON file
- `FIREBASE_PRIVATE_KEY` = From the downloaded JSON file (replace `\n` with actual newlines)
- `FIREBASE_CLIENT_EMAIL` = From the downloaded JSON file
- `FIREBASE_CLIENT_ID` = From the downloaded JSON file

Web Configuration:

- `FIREBASE_API_KEY` = Your Firebase API key (found in Project settings > General)
- `FIREBASE_AUTH_DOMAIN` = Your project ID + `.firebaseapp.com`
- `FIREBASE_STORAGE_BUCKET` = Your project ID + `.appspot.com`
- `FIREBASE_MESSAGING_SENDER_ID` = Found in Project settings
- `FIREBASE_APP_ID` = Found in Project settings

Step 3: Example Configuration

```
# Firebase Configuration
FIREBASE_PROJECT_ID=my-chicken-app-12345
FIREBASE_PRIVATE_KEY_ID=abc123def456
FIREBASE_PRIVATE_KEY="-----BEGIN PRIVATE KEY-----
\nMIIEvQIBADANBgkqhkiG9w0BAQEFAASCBAcwggSjAgEAAoIBAQC...\n-----END PRIVATE KEY-
----\n"
FIREBASE_CLIENT_EMAIL=my-app@my-chicken-app-12345.iam.gserviceaccount.com
FIREBASE_CLIENT_ID=123456789012345678901
FIREBASE_AUTH_URI=https://accounts.google.com/o/oauth2/auth
FIREBASE_TOKEN_URI=https://oauth2.googleapis.com/token
```

```
FIREBASE_AUTH_PROVIDER_X509_CERT_URL=https://www.googleapis.com/oauth2/v1/certs
FIREBASE_CLIENT_X509_CERT_URL=https://www.googleapis.com/robot/v1/metadata/x509
/my-app%40my-chicken-app-12345.iam.gserviceaccount.com

# Firebase Web/API Configuration
FIREBASE_API_KEY=AIzaSyDdI0hCZtE6vySjMm-WEfRq3CPzqKqqsHI
FIREBASE_AUTH_DOMAIN=my-chicken-app-12345.firebaseio.com
FIREBASE_DATABASE_URL=https://my-chicken-app-12345-default-rtdb.firebaseio.com
FIREBASE_STORAGE_BUCKET=my-chicken-app-12345.appspot.com
FIREBASE_MESSAGING_SENDER_ID=123456789012
FIREBASE_APP_ID=1:123456789012:web:abc123def456
FIREBASE_MEASUREMENT_ID=G-ABC123DEF4
```

Step 4: Restart Your Server

After updating the .env file:

1. Stop your server (Ctrl+C)
2. Restart it: `npm run dev`
3. Your Firebase integration should now work!

Troubleshooting

Common Issues:

1. **Invalid PEM formatted message:** Check your private key format
2. **Missing credentials:** Ensure all required fields are filled
3. **Permission errors:** Make sure your service account has proper Firebase permissions

Need Help?

- Check the Firebase documentation: <https://firebase.google.com/docs>
- Verify your service account has Editor permissions
- Ensure your Firebase project has the required APIs enabled

Security Notes

 **Important:** Never commit your `.env` file to version control!

- Add `.env` to your `.gitignore` file
- Keep your private keys secure
- Use different Firebase projects for development and production